

H524 Group Project - AI Comparison Analysis

Fall 2025 - Dr. John Molitor

Project Overview

This directory contains all materials for the H524 Introduction to Biostatistics Group Project. The project emphasizes **comparing AI tool outputs** (Claude, ChatGPT, Microsoft Copilot) while conducting statistical analysis on real medical datasets.

Key Innovation: Students don't just analyze data - they critically evaluate how different AI tools approach the same dataset, learning to identify strengths, weaknesses, and errors in AI-assisted statistical analysis.

Directory Structure

Group_project_John/

docs/	# Project documentation
final_project_guidelines.pdf	Complete project description
checkin.pdf	Week 4 data familiarization
DATA_DICTIONARIES.md	All 5 datasets documented
[LaTeX source files]	
group_1/	# Anorexia Treatment Study
anorexia.csv	Dataset (N=72)
README.md	Quick reference guide
group_2/	# Tooth Growth Study
toothgrowth.csv	Dataset (N=60)
README.md	Quick reference guide
group_3/	# Low Birth Weight Study
birthwt.csv	Dataset (N=189)
README.md	Quick reference guide
group_4/	# Pima Diabetes Study
pima_diabetes.csv	Dataset (N=200)
README.md	Quick reference guide
group_5/	# Diabetes Treatment Trial
diabetes2.csv	Dataset (N=699)
README.md	Quick reference guide
README.md	← You are here!

Project Timeline

Week	Milestone	Deliverable
Week 3	Team formation & dataset assignment	Teams formed
Week 4	Check-in (5% of grade)	Data familiarization report
Week 5-6	AI tool comparison & analysis	Working on project
Week 7	Optional draft for peer review	Draft (optional)
Week 8	Final submission (20% of grade)	Complete report & code
Week 9	Group presentations	15 min + 5 min Q&A

The Five Datasets

Quick Comparison Matrix

Group	Dataset	N	Design	Outcome	AI Divergence	Difficulty
1	Anorexia	72	Pre-post, 3 groups	Continuous	High	Medium
2	ToothGrowth	60	2×3 factorial	Continuous	High	Medium-High
3	Birth Weight	189	Case-control	Binary	Medium-High	Medium
4	Pima Diabetes	200	Cross-sectional	Binary	High	Medium (data cleaning!)
5	Diabetes Trial	699	RCT, 3 groups	Binary	Medium	Low (cleanest)

What “AI Divergence” Means

- **High ()**: AI tools will suggest different statistical methods, requiring critical evaluation
- **Medium ()**: Some differences in approach, but mostly agree
- **Low ()**: AI tools mostly agree (good for learning basics)

Higher divergence = more learning about AI comparison!

The AI Comparison Component (30% of grade!)

Three AI Tools to Compare

1. **Claude** (claude.ai)
 - Upload CSV file
 - Ask for statistical analysis suggestions
2. **ChatGPT** (chat.openai.com)
 - Upload CSV file
 - Ask for statistical analysis suggestions
3. **Microsoft Copilot** (in RStudio)
 - Use for R code generation
 - Works inline as you type

What to Compare

Students must document: - Statistical methods suggested by each AI - Assumption checks performed (or not!) - Effect sizes reported (or not!) - Code quality and clarity - Interpretation style - **Errors or omissions** - Which approach is best for intro biostatistics

Example Expected Differences

For Anorexia Dataset: - **Claude:** ANOVA on change scores, no assumption checks, accessible language - **ChatGPT:** ANCOVA adjusting for baseline, full assumptions, effect sizes - **Copilot:** Paired t-tests, code-focused

Students learn: How to critically evaluate AI outputs, not just accept them!

Key Documents

For Students

1. **Start here:** docs/final_project_guidelines.pdf
 - Complete project description
 - Grading rubric
 - Timeline and deliverables
2. **Check-in:** docs/checkin.pdf
 - Data familiarization assignment
 - Team meeting requirements
 - Getting started guide (use AI for R help!)
3. **Dataset information:** docs/DATA_DICTIONARIES.md
 - Complete variable descriptions
 - Research context
 - Suggested methods
4. **Your group folder:** group_X/README.md
 - Quick reference for your dataset
 - Loading instructions
 - Expected AI differences

For Instructor

All materials are ready to distribute: - PDFs can be uploaded to Canvas - CSV files in group folders ready to share - READMEs provide quick reference for students - LaTeX source available for future edits

What Makes This Project Different?

Traditional Biostatistics Project:

1. Load data
2. Run statistical tests
3. Write report
4. Done

This AI-Enhanced Project:

1. Load data
2. **Consult three different AI tools**
3. **Compare AI suggestions critically**

4. **Verify AI outputs** (check assumptions, calculations)
5. **Synthesize best approach** from multiple sources
6. Document AI comparison process
7. Write report explaining what you learned about AI

Result: Students learn to work WITH AI, not just blindly use it.

Pedagogical Features

Why These 5 Datasets?

Each dataset was chosen to elicit **different AI responses**:

1. **Anorexia** - Pre-post design creates method ambiguity
2. **ToothGrowth** - Factorial design, some AI suggest advanced methods
3. **Birth Weight** - Binary outcome, multiple predictors, method choices
4. **Pima Diabetes** - **Hidden data quality issues!** Tests AI data cleaning
5. **Diabetes Trial** - Cleanest dataset, “control” for comparison

Learning Objectives

Students learn: - Statistical methods (ANOVA, t-tests, chi-square) - Assumption checking (normality, equal variances) - Data cleaning and quality assessment - **Critical AI evaluation** - **AI verification strategies** - **When AI is helpful vs. when it makes mistakes**

Dataset Sources

All datasets from **MedDataSets R package** (Caceres Rossi, 2024): - Anorexia: Hand et al. (1994) - ToothGrowth: Crampton (1947) - Birth Weight: Hosmer & Lemeshow (1989) - Pima Diabetes: NIDDK study - Diabetes Trial: Clinical trial data

Advantage: Real medical data, vetted for teaching, intro-appropriate complexity

Getting Started (For Students)

Step 1: Read the Guidelines

Open docs/final_project_guidelines.pdf

Step 2: Find Your Group Folder

Go to group_X/ where X is your team number (1-5)

Step 3: Read Your Dataset README

Open group_X/README.md for quick overview

Step 4: Check-In

Follow docs/checkin.pdf instructions

Step 5: Consult AI Tools

Upload your CSV to Claude, ChatGPT, and use Copilot in RStudio

Step 6: Compare and Analyze

Document what each AI suggested and evaluate critically

Technical Notes

Data Loading Options

Option 1: CSV files (provided in group folders)

```
data <- read.csv("your_dataset.csv")
```

Option 2: MedDataSets package (optional)

```
library(MedDataSets)
data("dataset_name_df")
```

Both work! CSV files ensure everyone can start immediately.

Required R Packages

```
library(tidyverse)    # Data manipulation
library(ggplot2)      # Visualization
library(stats)        # Statistical tests
```

Getting Help

- **Office Hours:** Wednesday 10:00-11:30 AM
 - **Canvas Discussion Board:** Post questions about datasets or methods
 - **Email:** For team scheduling or private concerns
 - **Special Office Hours:** Weeks 5, 6, 7 (watch for announcements)
-

Completion Checklist

All materials ready: - [x] 5 datasets extracted and placed in group folders - [x] Final project guidelines (PDF)
- [x] Check-in guidelines (PDF) - [x] Comprehensive data dictionaries - [x] Individual README files for each group - [x] LaTeX source files for future edits - [x] Main overview README (this file)

Status: Ready to distribute to students!

Version History

- **November 2, 2025:** Initial creation
 - Complete project structure built
 - All 5 datasets extracted from MedDataSets
 - Documentation created (guidelines, dictionaries, READMEs)
 - LaTeX compiled to PDF
 - Tested with R to verify datasets load correctly
-

Summary

This project teaches students to: 1. Conduct proper statistical analysis 2. **Use AI tools effectively** 3. **Critically evaluate AI outputs** 4. **Verify and synthesize information** 5. Work in teams 6. Communicate findings professionally

Bottom line: Preparing students for the reality of modern statistical practice where AI is a powerful assistant, but human oversight is essential.

Created by: Claude Code **Date:** November 2, 2025 **For:** H524 Introduction to Biostatistics, Fall 2025
Instructor: Dr. John Molitor